

CHE 311

Hw # 11

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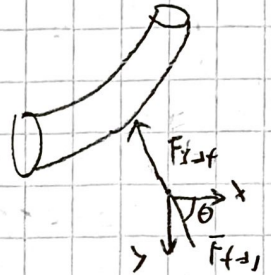
Allen choi

UIN: 665774571

$$\angle \theta = 45^\circ \quad P_1 = 21 \text{ psi} \quad P_2 = 19 \text{ psi} \quad Q = 3 \text{ ft}^3/\text{s} \quad \rho = 62.4 \text{ lbm/ft}^3 \quad \mu = 10 \text{ cp} = 6.72 \times 10^{-4} \text{ lbm/ft-s}$$

$$L = 1 \text{ ft} \quad R = 4 \text{ in}$$

$$\vec{F}_{f \rightarrow s} = -\vec{F}_{s \rightarrow f} = P_1 \langle v^2 \rangle S_1 \vec{u}_1 - P_2 \langle v^2 \rangle S_2 \vec{u}_2 + P_1 S_1 \vec{u}_1 - P_2 S_2 \vec{u}_2 + m\vec{g}$$



$$\vec{F}_{f \rightarrow s} = \rho \langle v^2 \rangle S - \rho \langle v^2 \rangle S \cos 45^\circ + P_1 S - P_2 S \cos 45^\circ$$

$$F_{f \rightarrow s} = 0 - \rho \langle v^2 \rangle S \sin 45^\circ + 0 - P_2 S \sin 45^\circ - mg = -\rho \langle v^2 \rangle S \sin 45^\circ - P_2 S \sin 45^\circ - mg$$

$$\langle v \rangle = \frac{Q}{A} = \frac{3 \text{ ft}^3/\text{s}}{\pi (4 \text{ in})^2 \times \frac{1 \text{ ft}}{12 \text{ in}}} = 8.594 \text{ ft/s}$$

$$Re = \frac{\rho v D}{\mu} = \frac{62.4 \text{ lbm/ft}^3 \times 8.594 \text{ ft/s} \times 8 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}}}{6.72 \times 10^{-4} \text{ lbm/ft-s}} = 5320/0$$

Turbulent flow, so $\langle v \rangle^2 = \langle v^2 \rangle$ X-axis:

$$F_{f \rightarrow s, x} = \rho \langle v^2 \rangle S (1 - \cos 45^\circ) + S (P_1 - P_2 \cos 45^\circ)$$

$$= 62.4 \text{ lbm/ft}^3 \cdot (8.594 \text{ ft/s})^2 \cdot (4 \text{ in})^2 \pi \times \frac{1 \text{ ft}}{(12 \text{ in})^2} (1 - 0.7071) + (4 \text{ in})^2 \times \pi \times \frac{1 \text{ ft}}{(12 \text{ in})^2} \left(21 \frac{\text{lb}}{\text{in}^2} - 19 \frac{\text{lb}}{\text{in}^2} \cdot 0.7071 \right)$$

$$= 471.20 \text{ lbm ft/s}^2 \times \frac{1 \text{ lbf}}{32.174 \text{ lbm ft/s}^2} + 2.641 \frac{\text{ft}^2 \text{ lbf}}{\text{in}^2} \times \frac{(12 \text{ in})^2}{1 \text{ ft}^2} \cdot 0.7071$$

$$= 14.645 \text{ lbf} + 380.304 \text{ lbf} = 394.949 \text{ lbf}$$

Y-axis:

$$F_{f \rightarrow s, y} = -\rho \langle v^2 \rangle S \sin 45^\circ - P_2 S \sin 45^\circ - mg$$

$$= -62.4 \text{ lbm/ft}^3 \times (8.594 \text{ ft/s})^2 \cdot \pi \cdot (4 \text{ in})^2 \times \frac{1 \text{ ft}}{(12 \text{ in})^2} \cdot 0.7071 - 19 \frac{\text{lb}}{\text{in}^2} \times (4 \text{ in})^2 \cdot \pi \cdot 0.7071$$

$$= -1137.532 \text{ lbm ft/s}^2 \times \frac{1 \text{ lbf}}{32.174 \text{ lbm ft/s}^2} - 675.312 \text{ lbf} - 697.015 \text{ lbm ft/s}^2 \times \frac{1 \text{ lbf}}{32.174 \text{ lbm ft/s}^2}$$

$$= -732.332 \text{ lbf}$$

$$|F_{f \rightarrow s}| = \sqrt{(F_{f \rightarrow s, x})^2 + (F_{f \rightarrow s, y})^2} = 832.043 \text{ lbf}, \quad \theta = \tan^{-1} \left(\frac{F_{f \rightarrow s, y}}{F_{f \rightarrow s, x}} \right) = 61.662^\circ$$

2. $\rho = 62.4 \text{ lbm/ft}^3$, $\mu = 1 \text{ cp}$ $Q = 3 \text{ ft}^3/\text{s}$ turbulent flow

$D = 4 \text{ in}$ $P_1 = 21 \text{ psia}$ $P_2 = 19 \text{ psia}$

$$S = A = \pi R^2 = \pi (2 \text{ in})^2 = 4\pi \text{ in}^2 \times \frac{1 \text{ ft}^2}{(12 \text{ in})^2} = \frac{\pi}{36} \text{ ft}^2$$

$$Q = A \langle v \rangle$$

$$\Rightarrow \langle v \rangle = \frac{Q}{A} = \frac{3 \text{ ft}^3/\text{s}}{\pi/36 \text{ ft}^2} = 34.377 \text{ ft/s}$$

X-axis: $F_{f_{x,x}} = \rho \langle v \rangle^2 S + \rho \langle v \rangle^2 S + P_1 S + P_2 S$

$$= 2 \times (34.377 \text{ ft/s})^2 \times 62.4 \frac{\text{lbm}}{\text{ft}^3} \cdot \frac{\pi}{36} \text{ ft}^2 + \frac{\pi}{36} \text{ ft}^2 \cdot (21 + 19) \frac{\text{lbf}}{\text{in}^2} \times \frac{144 \text{ in}^2}{1 \text{ ft}^2}$$

$$= 12370.374 \text{ lbm ft/s}^2 \times \frac{1 \text{ lbf}}{32.174 \text{ lbm ft/s}^2} + 502.655 \text{ lbf}$$

$$= 902.635 \text{ lbf}$$